

KVAMSOY, Norway

WMO: 013290

Lat: 60.364N

Lon: 6.279E

Elev: 164

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	24.8	8.8	8.7	26.1	11.7	10.0	28.0	43.7	37.6	40.1	35.8	14.1	50	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	77.4	63.7	73.9	61.6	70.7	60.3	65.3	75.0	63.3	71.4	61.7	68.8	6.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	81.7	69.2	59.8	77.2	67.3	58.3	73.2	65.4	30.2	75.3	28.8	71.4	27.6	68.8	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.5	29.5	25.8	DB	21.2	82.5	3.3	3.6	18.8	85.1	16.9	87.2	15.0	89.3	12.6	91.9	
			WB	18.7	67.4	3.6	2.5	16.1	69.2	14.0	70.7	12.0	72.0	9.3	73.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	34.9	35.1	38.4	45.2	52.6	58.3	62.6	60.4	54.9	47.5	41.1	36.7
	DBStd	11.08	4.61	4.96	4.91	5.17	6.48	5.26	5.81	4.49	4.74	4.59	5.47	5.48
	HDD50	2245	468	418	359	161	44	3	0	0	10	103	268	411
	HDD65	6499	933	838	824	593	388	212	118	155	303	543	716	876
	CDD50	1294	0	0	0	18	125	251	391	323	158	25	3	0
	CDD65	72	0	0	0	0	4	10	44	13	1	0	0	0
	CDH74	303	0	0	0	0	23	46	208	25	1	0	0	0
	CDH80	45	0	0	0	0	4	3	37	1	0	0	0	0
(14) Wind	WSAvg	9.6	12.8	10.3	10.9	8.6	8.0	7.8	7.3	7.1	8.4	10.3	11.6	11.6
(15) Precipitation	PrecAvg	88.9	10.3	8.6	8.5	4.4	3.9	4.7	5.1	5.8	8.2	9.7	9.0	10.6
	PrecMax	128.1	19.6	18.7	27.5	8.9	8.5	9.5	7.9	9.8	14.3	20.8	18.6	18.9
	PrecMin	55.7	0.1	0.0	1.1	1.3	0.6	1.5	1.6	2.5	2.4	2.7	1.7	1.4
	PrecStd	15.2	5.5	5.2	5.3	1.9	2.0	2.0	1.5	2.0	3.7	4.4	4.6	5.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	47.4	53.4	63.5	77.3	78.9	84.3	77.5	69.4	62.4	55.3	49.6
		MCWB	42.0	41.8	43.7	50.8	61.8	62.5	66.4	63.8	58.9	54.2	48.8	44.9
	2%	DB	44.6	45.4	49.4	59.7	70.6	74.6	79.5	72.9	66.4	58.2	52.4	47.2
		MCWB	41.6	41.1	42.1	48.8	57.3	59.8	65.6	61.9	57.6	51.4	47.0	43.5
	5%	DB	42.7	43.7	46.8	56.5	66.8	71.0	75.6	70.0	63.8	55.6	50.8	45.5
		MCWB	39.2	39.7	41.2	47.1	54.9	58.4	63.7	60.5	56.3	50.5	46.4	42.3
	10%	DB	41.0	41.9	44.8	53.7	62.8	67.4	71.9	67.5	61.4	53.5	48.9	43.9
		MCWB	38.0	38.4	40.6	45.3	52.7	57.0	62.3	59.4	55.2	48.8	45.5	40.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	44.4	45.5	52.2	62.0	63.9	69.0	65.6	60.8	55.6	50.5	47.2
		MCDB	45.3	46.0	48.6	61.8	75.0	75.9	80.7	74.6	66.1	58.6	52.4	48.1
	2%	WB	42.4	42.6	43.9	49.7	58.4	61.2	66.5	63.3	59.0	53.7	49.0	45.0
		MCDB	43.8	44.5	47.2	57.1	69.4	71.4	77.1	70.6	63.8	57.2	51.4	46.4
	5%	WB	40.3	40.8	42.6	48.2	55.8	59.4	64.7	61.8	57.7	51.8	47.5	43.0
		MCDB	42.1	42.9	45.7	55.2	65.1	68.9	73.5	67.8	62.2	54.5	49.9	44.8
	10%	WB	38.6	39.1	41.1	46.3	53.6	57.8	63.0	60.5	56.3	50.0	45.5	41.2
		MCDB	40.6	41.3	44.6	52.4	61.4	65.9	70.7	65.9	60.4	52.7	48.1	43.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.1	8.1	11.5	13.8	13.7	12.9	11.1	8.8	6.8	5.8	5.3
		MCDBR	7.8	7.8	10.5	15.2	19.4	20.1	19.4	15.4	12.3	9.0	7.8	7.7
	5% WB	MCWBR	6.2	5.6	6.5	7.9	10.0	10.0	8.5	7.5	6.4	5.3	5.6	6.7
		MCWBR	7.5	7.5	9.1	14.3	18.5	17.7	17.7	13.9	10.8	7.4	7.4	7.6
(40) Clear-Sky Solar Irradiance	taub		0.261	0.300	0.335	0.372	0.374	0.368	0.382	0.376	0.348	0.325	0.279	0.239
		taud	2.091	2.173	2.154	2.133	2.303	2.383	2.373	2.386	2.454	2.400	2.189	2.018
	Ebn at Noon		153	216	246	259	270	273	266	258	246	210	147	108
		Edh at Noon	14	23	32	40	36	34	34	31	24	19	12	9
(44) All-Sky Solar Radiation	RadAvg		81	250	642	1211	1526	1556	1456	1133	714	335	113	53
	RadStd		19	64	121	192	239	187	174	103	83	49	18	7

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page